

MAE 4730

9/17/18

Lecture 10

Q&A for 10b:

- Can use big perturbation ($\Delta r + C$) not ($\Delta r + \omega$)
- `fminsearch` (IC, period)
 - IC match final condition
 - Initial position + velocity needs to match for it to be periodic
- `fsolve` (IC, period)
 - Try to make IC + FC same
- `fmincon`

Back to falling cat:

- No theorem saying internal forces can't turn
 - Bad bearing - torque opposing motion $\rightarrow$ turn the other way
- Good bearing $\rightarrow$ $\rightarrow$ continues turning
- Accelerating point  no conservation of angular momentum about an accelerating point

